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Establishment of cell-based transposon-mediated transgenesis in cattle

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Transposon-mediated transgenesis is a well-established tool for genome modification in small animal models. However, translation of this active transgenic method to large animals warrants further investigations. Here, the piggyBac (PB) and sleeping beauty (SB) transposon systems were assessed for stable gene transfer into the cattle genome. Bovine fibroblasts were transfected either with a helper-independent PB system or a binary SB system. Both transposons were highly active in bovine cells increasing the efficiency of DNA integration up to 88 times over basal nonfacilitated integrations in a colony formation assay. SB transposase catalyzed multiplex transgene integrations in fibroblast cells transfected with the helper vector and two donor vectors carrying different transgenes (fluorophore and neomycin resistance). Stably transfected fibroblasts were used for SCNT and on *in vitro* embryo culture, morphologically normal blastocysts that expressed the fluorophore were obtained with both transposon systems. The data indicate that transposition is a feasible approach for genetic engineering in the cattle genome.

1. Introduction

The advent of high throughput DNA sequencing methods and comprehensive annotated genome maps concomitantly with advanced active transgenic techniques promise to revolutionize the field of animal biotechnology. In particular, areas like disease modeling, biopharming, and basic research will benefit enormously by introducing precise genetic engineering tools to manipulate livestock genomes. Initial transgenic methods relayed on passive (nonfacilitated) genomic integration of transgenes at sites of spontaneously arising double-strand breaks of chromosomes after direct injection of naked DNA into zygotes (pronuclear injection) or transfection of cultured cells followed by SCNT. Homologous recombination in somatic cells of livestock is an extremely rare event, and only a few

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